

AECCS Final Presentation

Automated GDPR compliance analysis and PET evaluation for cookie consent systems.

CS475 Privacy-Enhancing Technologies

Research Question

- Do cookie consent systems actually protect users from non-essential tracking?
- How often do banners rely on dark patterns?
- Which PETs reduce exposure most effectively when consent systems fail?

What Was Built

- Crawler with no-interaction, accept, and reject states.
- Tracker classification using filter lists and heuristics.
- Dark-pattern detection and GDPR-oriented compliance scoring.
- Browser PET evaluation, CMP comparison, and differential privacy reporting.
- Visualization and HTML reporting pipeline with provenance checks.

Current Repository Status

Area	Status
Software pipeline	Implemented
Mock/demo dataset	Reproducible
Real study list	Prepared with 100 domains
Real crawl outputs	Completed (97/100 sites, 1 March 2026)
Final empirical results	Completed (100-site study with 7 PETs)

Real-Study Findings (100 Sites, 1 March 2026)

- 89.7% of sites set pre-consent trackers before any user interaction.
- 82.0% of sites exhibit at least one dark pattern (most common: missing reject button at 69%).
- Average compliance score is 33.1 / 100 (median 24.0); 67% of sites received grade F.
- Brave Shields is the strongest browser PET (14.7% average tracker reduction).
- OneTrust is the strongest CMP as a privacy tool (PET score 33.4).
- Best combination: Brave Shields + OneTrust (estimated 43.2% effectiveness).
- Recommended DP epsilon: 1.0 for balanced privacy/utility when publishing aggregate statistics.

Based on real crawl data from data/real/, run ID real-study-20260301-final.

Completed Actions

- Real crawl and PET evaluation completed under data/real/ (1 March 2026).
- All 11 figures and HTML compliance report regenerated from real processed data.
- Demo findings replaced with real-study results across all documents.